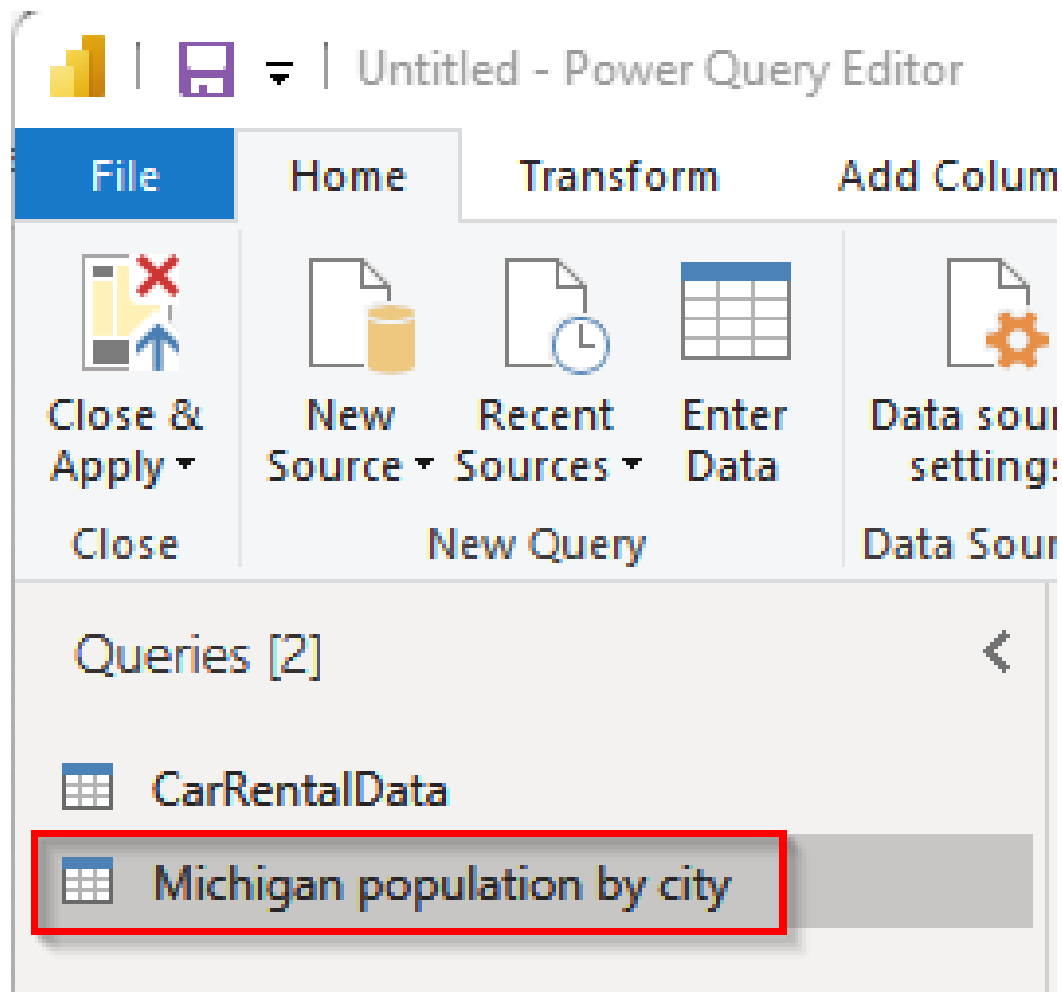


Assignment 4 - Instructions

(1) Datasets

1. Download the [Cornell Car Rental Dataset](#) from Kaggle (File name is: CarRentalData.csv) and import it into Power BI. You might want to register for an account before downloading. Take a quick look at the distributions, data format and values of variables on their web site.
2. Import the populations of Michigan, Ohio and Illinois cities from **World Population Review**. This will require you to perform vertical merging of data sets. Follow the steps below:
 - a. Click Home | Transform data in Power BI. This will take you to Power Query.
 - b. In Power Query, click Home | New Source to retrieve [Michigan population by city](#). Rename the query to **Michigan population by city** as shown in the screenshot below.



- c. Create a custom column called **State** that stores “Michigan” as the values. Why? This column distinguishes a state from another after data merging.

Custom Column

Add a column that is computed from the other columns.

New column name

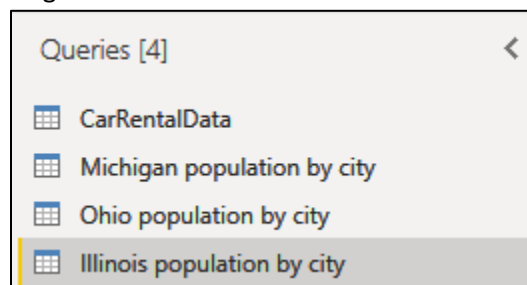
State

Custom column formula ⓘ

= "MI"

1.2	Area (mi²)	State
4500	138.72	MI
4493	44.76	MI
4088	34.36	MI
3713	36.43	MI
4508	27.91	MI
4525	24.23	MI

- d. Click the **Keep Rows** button to keep only the top 100 ranked cities.
- e. Now repeat steps b, c and d to import populations of Ohio and Illinois cities. Don’t forget to create the **State** column for each. You should now have the following queries.



- f. Since vertical merging requires that all data sets have the same number of columns and the same column names, the above steps in c, d, and e have already prepared us ready for it.

Now, append all three population queries into one by using **Append queries as new**. Name this new query **Population by city – MI, OH, IL**. Double-check if data includes all three states. Send the data back to Power BI.

(2) Data Manipulation

Do all the data manipulation steps in Power BI, not in Power Query.

1. Since there may be cities with the same name between states, let's create a **city.state** calculated column in **CarRentalData** and **Population by city – MI, OH, IL** tables in Power BI (not in Power Query). The column will show something like **Kalamazoo, MI**. Don't forget to configure Power BI so that it knows that this is a location.
2. Create a relationship between **CarRentalData** and **Population by city – MI, OH, IL** using the newly created **city.state** column. This should be a one-to-many relationship.
3. Create a calculated column called **new.vehicle.type** to combine minivan and van categories into one category called **van_minivan**. All other categories remain the same. Use nested IF or IF statement for this.
4. Create a calculated column called **rating category** with only the following four categories. This technique is called discretization that converts numeric values in a column into categories. We talked about how to do it in Power Query. Here we are doing it in Power BI.
 - **No rating** if the rating column has no value.
 - **Low rating** if the ratings are less than or equal to 2.
 - **Medium rating** if the ratings are great than 2, but less than 4.
 - **High rating** if the ratings are greater than or equal to 4.

The following DAX Nested IF Statement can Create the column for you. Make sure to understand how it works.

```
rating category = IF(ISBLANK(CarRentalData[rating]),"No rating",  
IF(CarRentalData[rating] <= 2, "Low rating", IF(CarRentalData[rating] < 4, "Medium  
rating", "High rating")))
```

- ✓ You can also use "**SWITCH**" DAX function here instead of Nested IF. [Here](#) you can learn more about SWITCH function.
5. Create a measure called **Max number of reviews** to calculate the maximum number of reviews.
 6. Create a measure called **Average Trips taken – Recent vehicle year** to calculate the average renter trips taken for vehicle years after 2015. You might use CALCULATE DAX Function. [Here](#) you can learn more about CALCULATE function.
 7. Create another measure called **Average Trips taken – Older vehicle year** for vehicle years 2015 or before. You might use CALCULATE DAX Function. [Here](#) you can learn more about CALCULATE function.

(3) Visuals:

1. Create a card visual for each of the following. (Note that the first one uses a measure. The other three use normal columns. This shows that you can either use a measure or a column for card visuals.)
 - a. Maximum number of reviews (use the Max number of reviews measure for this)
 - b. Average rating
 - c. Average population density
 - d. Total population

You can use one single card for each of them. Your card visuals should look like the following (text, style, color and shape). There is a shadow for each card. The year could be 2023 or later depends on the year you download the population data.



2. Create a Map visual to show car rental locations.
 - a. Select **city.state** from the **Population by city** table that you created above. (Why not city.state from the CarRentalData table?)
 - b. Define an explicit filter to select only MI, OH and IL for the page (not for the visual or for all pages).
 - c. Drag **new vehicle type** into the Legend well, reviewCount into the Size well, and other vehicle information (average rating, average daily rate, and average year) to the Tooltips well.
 - d. You can change the Bobble size in the Format Pane.
3. Create a key influencer visual to analyze Average rating by the following columns:
 - a. New vehicle type
 - b. Average vehicle year
 - c. Average 2023 population
 - d. Average population density
 - e. Average daily rate
 - f. Maximum number of reviews
4. Create two slicer visuals, one for the new vehicle type and the other for the rental company's states. See the style in the screenshot below.
5. Create a matrix visual to show **city.state** and **new vehicle type** in Rows, rating category in Columns and average daily rate in Values.
6. Arrange your visuals in the following way. Pay attention to the title, style, shape and color of the visuals.
7. Change the Name of the report from Page 1 to "Car Rental".

